

**ОТЧЕТ**  
**по лабораторной работе №3**  
**САРАНЦЕВОЙ ДАРЬИ**  
**ВАРИАНТ 18**

All tasks:

task 1: Displays comparison between Taylor series calculation and `math.asin()` result.

task 2: Sum the last digits of entered numbers until stop value is entered.

task 3: Count the number of digits and vowels in a string.

task 4: Analyzes a fixed text.

task 5: Find max absolute element and sum of elements before the last even element.

Please, select task you want to check: 1

Input x value: 0/9

Invalid input. Please enter a number between -1 and 1.

Input x value: 0.9

Input accuracy of calculations: 1e-2

| x         | n | F(x)        | Math F(x)   | eps      |
|-----------|---|-------------|-------------|----------|
| +0.900000 | 6 | +1.10593050 | +1.11976951 | 1.00e-02 |

→ Terms used: 6

→ Error: 1.38e-02

Do you want to try task again?(y/n): n

Do you want to continue?(y/n): y

All tasks:

task 1: Displays comparison between Taylor series calculation and `math.asin()` result.

task 2: Sum the last digits of entered numbers until stop value is entered.

task 3: Count the number of digits and vowels in a string.

task 4: Analyzes a fixed text.

task 5: Find max absolute element and sum of elements before the last even element.

Please, select task you want to check: 2

Enter integer (enter 18 to stop): 2

Last digit: 2, Total: 2

Enter integer (enter 18 to stop): 456

Last digit: 6, Total: 8

Enter integer (enter 18 to stop): 59

Last digit: 9, Total: 17

Enter integer (enter 18 to stop): 1110

Last digit: 0, Total: 17

Enter integer (enter 18 to stop): 18

Sum result: 17

Do you want to try task again?(y/n): █

```

Please, select task you want to check: 3
Print your string: argjkdf djjs      3ggg i
Calling calculate_numbers_and_vowels with args: ('argjkdf djjs      3ggg i',)
calculate_numbers_and_vowels returned: (2, 2)
wrapper executed in 0.000936 seconds
In string 2 numbers and 2 vowels
Do you want to try task again?(y/n): n
Do you want to continue?(y/n): y
All tasks:
task 1: Displays comparison between Taylor series calculation and math.asin() result.
task 2: Sum the last digits of entered numbers until stop value is entered.
task 3: Count the number of digits and vowels in a string.
task 4: Analyzes a fixed text.
task 5: Find max absolute element and sum of elements before the last even element.
Please, select task you want to check: 4
a) Words with 4 letters: 9
['mind', 'well', 'made', 'feel', 'very', 'when', 'with', 'pink', 'eyes']

b) Words with equal vowels and consonants:
1: so
5: in
9: as
11: as
18: made
20: feel
21: very
22: sleepy
27: pleasure
28: of
34: be
38: of
40: up
55: by

c) Words sorted by length descending:
['considering', 'pleasure', 'suddenly', 'whether', 'trouble', 'getting', 'picking', 'daisies', 'sleepy', 'stupid', 'making', 'Rabbit', 'could', 'daisy', 'chain', 'would', 'worth', 'white', 'close', 'mind', 'well', 'made', 'feel', 'very', 'when', 'with', 'pink', 'eyes', 'she', 'was', 'her', 'own', 'she', 'for', 'the', 'hot', 'day', 'her', 'and', 'the', 'the', 'and', 'the', 'ran', 'her', 'so', 'in', 'as', 'as', 'of', 'be', 'of', 'up', 'by', 'a', 'a']
Do you want to continue?(y/n): 

```

```

Please, select task you want to check: 5
Enter list size (1-20): 5
Do you want do generate list?(y/n): y
Enter minimum value: -5
Enter maximum value: 12

```

Generated list: [5, 1, 5, 11, 10]

Original list: [5, 1, 5, 11, 10]

Results:

Maximum by absolute value: 11 (at index 3)

Sum of elements before last even: 22

Do you want to process another list?(y/n): y

Enter list size (1-20): 5

Do you want do generate list?(y/n): n

Enter 5 integer(s):

Element 1: -100

Element 2: 2

Element 3: sdf

Invalid input. Please enter an integer.

Element 3: 6

Element 4: 0

Element 5: 2

Original list: [-100, 2, 6, 0, 2]

Results:

Maximum by absolute value: -100 (at index 0)

Sum of elements before last even: -92